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A fractional study based on the economic and environmental mathematical model

Qiliang Chen^a, Zulqurnain Sabir^b, Muhammad Asif Zahoor Raja^c, Wei Gao^d,
Haci Mehmet Baskonus^{e,*}

^a School of Economics and Finance, Chongqing University of Technology, Chongqing, China

^b Department of Mathematics and Statistics, Hazara University, Mansehra, Pakistan

^c Future Technology Research Center, National Yunlin University of Science and Technology, 123 University Road, Section 3, Douliou, Yunlin 64002, Taiwan, ROC

^d School of Information Science and Technology, Yunnan Normal University, Kunming 650500, China

^e Faculty of Education, Harran University, Sanliurfa 63050, Turkey

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Abstract The purpose of this investigation is to provide the numerical study based on the fractional order (FO) economic and environmental mathematical model (EEMM) called as FO-EEMM. The motive of this work is to find more realistic results of the EEMM of the non-integer and FO derivatives. The structure of the FO-EEMM is categorized into three dynamics, control accomplishment cost, capability of the manufacturing elements and the diagnostics cost of the technical exclusion. The solution of the FO-EEMM is numerically presented by using the scaled conjugate gradient neural networks (SCGNNs). Three cases based on the FO-EEMM have been scrutinized to indicate the numerical performances along with the selection of the statics as 76% for training, 13% for testing and 11% for certification. The correctness of the designed SCGNNs is authenticated by using the matching of the achieved and the reference solutions (Adams-Bashforth-Moulton). The validity, exactness, dependability, and competence of the SCGNNs is observed through the performances of the mean square error, regression, state transitions, error histograms and correlation.

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1. Introduction

In recent years, scientists have found the study of economics and the environment to be one of the more fascinating fields of research because of the range of industrial applications. By improving financial results and operational organizations as well as by lowering the overall cost and catalogs in the

* Corresponding author.

E-mail addresses: chenqiliang123@163.com (Q. Chen), zulqurnain_maths@hu.edu.pk (Z. Sabir), rajamaz@yuntech.edu.tw (M. Asif Zahoor Raja), gaowei@ynnu.edu.cn (W. Gao), hmbaskonus@harran.edu.tr (H. Mehmet Baskonus).

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process of the supply chain organization (SCO), theoretical studies of the connotation of the supply chain are used to the strategic associations [1–3]. The connecting nature of two or more firms in the process of the SCO has been reported by Mentzer et al [4]. The categories of the industrial growing quantity based on the strategic importance of planning and controlling together with the SCO design. The relationship with industrial organizations is used to strengthen cultural activities and generate greater benefits [5,6]. Min et al [7] worked on the SCO modeling procedures by operating the key challenges along with the supply chain model opportunities. Table 1.

For the prediction of the integrated economy growth, the growing ratio of the low-level of economies become unstable, when the advanced countries of the world face economic disaster. The model based on the construction of the “predator-victim” represent the validations of the mathematical systems. The mathematical formulations of the fractional order (FO) economic and environmental mathematical model (EEMM) called as FO-EEMM are categorized into three dynamics, control accomplishment cost, capability of the manufacturing elements and the diagnostics cost of the technical exclusion. The general form of the integer order EEMM is given as [8–10]:

$$\begin{cases} \frac{du(\varepsilon)}{d\varepsilon} = C_3w(\varepsilon) + C_1u(\varepsilon)(a - u(\varepsilon)) - C_2v(\varepsilon), u_0 = k_1, \\ \frac{dv(\varepsilon)}{d\varepsilon} = C_4u(\varepsilon)(a - u(\varepsilon)) + C_6w(\varepsilon) - C_5v(\varepsilon)(b - v(\varepsilon)), v_0 = k_2, \\ \frac{dw(\varepsilon)}{d\varepsilon} = C_7u(\varepsilon) - C_8v(\varepsilon), w_0 = k_3, \end{cases} \quad (1)$$

where, $C_1, C_2, C_3, C_4, C_5, C_6, C_7$ and C_8 represent the coefficient constants. k_1, k_2, k_3 are the initial conditions. On the basis of the FO-EEMM, the system above has been upgraded and is displayed as:

$$\begin{cases} \frac{d^\rho u(\varepsilon)}{d\varepsilon^\rho} = C_3w(\varepsilon) + C_1u(\varepsilon)(a - u(\varepsilon)) - C_2v(\varepsilon), u_0 = k_1, \\ \frac{d^\rho v(\varepsilon)}{d\varepsilon^\rho} = C_4u(\varepsilon)(a - u(\varepsilon)) + C_6w(\varepsilon) - C_5v(\varepsilon)(b - v(\varepsilon)), v_0 = k_2, \\ \frac{d^\rho w(\varepsilon)}{d\varepsilon^\rho} = C_7u(\varepsilon) - C_8v(\varepsilon), w_0 = k_3, \end{cases} \quad (2)$$

where ρ represents the fractional Caputo derivative. The values of the ρ have been taken between 0 and 1. Performing precise and realistic solutions is the aim behind using fractional derivatives. In FO models, minute characteristics based on the superfast transition and super slow evolution are explored, which gives more detail of the system dynamics by using the fractional calculus but is difficult to explain by using the integer order counterparts. The fractional calculus is additionally used to perform the system dynamics. The FO derivatives perform significantly better than the integer order. The FO derivatives have been applied in the variety of applications, like as fractional integro mixed immunotherapy and chemotherapy

cancer treatment fractional model [11], diffusion equation [12], virus mathematical systems [13–16], beam motion on nanowire [17], capacitor microphone [18], nonlinear models [19–21], quadratic nonlinear equations [22], SIR epidemic system of child disease [23], atmospheric dynamics of carbon dioxide gas [24] and Lakshmanan–Porsezian–Daniel system [25]. Additionally, a variety of applications based on control networks, physical and engineering have been intensively researched using FO derivatives. There are various operators that have been used in the fractional calculus, e.g., Grünwald-Letnikov, Weyl-Riesz, Riemann-Liouville, Caputo and Erdlyi-Kober. Each of these operators is valuable and important in their own sense. However, the definition of the Caputo derivative that is most frequently used to solve both homogeneous and non-homogeneous initial conditions. Compared to the other definitions, the Caputo derivatives are considered to be easier to implement. The authors are motivated to develop the FO-EEMM based on these FO applications and provide the numerical performances using scaled conjugate gradient neural networks (SCGNNs).

The remaining Sections of this study are presented as: Novel features along with the applications of the stochastic solvers are reported in the 2nd Section. The stochastic SCGNNs is given in the 3rd Section. The results based on the FO-EEMM are provided in Section 4. Conclusions along with the future research directions are shown in final Section.

2. Novel characteristics and outline of the stochastic computing solvers

In this Section, the SCGNNs stochastic solvers have been presented for the FO-EEMM. The stochastic computing solvers have been verified with the local as well as global search operator presentations for the number of singular and biological models [26]. Few more applications of the stochastic solvers are nervous stomach TFM system [27]. Some novel topographies of the SCGNNs for solving the FO-EEMM are given as:

- The mathematical formulation of the nonlinear EEMM is extended into the FO derivatives.
- The numerical stochastic procedures have never been implemented before to present the solutions of the mathematical FO-EEMM.
- The numerical performances of the mathematical FO-EEMM have been effectively described by applying the stochastic performances.
- Three variations of the fractional kind of derivatives have been scrutinized to indicate the numerical measures of the mathematical FO-EEMM.

Table 1 SCGNNs measures for the FO-EEMM.

Case	MSE	Gradient	Performance	Iteration	Mu	Time		
	Authentication	Training	Testing					
1	9.786×10^{-10}	9.330×10^{-10}	2.656×10^{-10}	1.00×10^{-07}	9.33×10^{-10}	864	1×10^{-09}	5
2	3.288×10^{-09}	5.704×10^{-10}	8.051×10^{-09}	9.97×10^{-08}	5.70×10^{-10}	500	1×10^{-09}	4
3	1.405×10^{-12}	2.468×10^{-12}	5.503×10^{-12}	9.64×10^{-08}	2.46×10^{-12}	515	1×10^{-09}	4

1. Problem: Nonlinear FO-EEMM

Computational intelligent framework

A procedure through the multi-layer nets is programmatic using the SCGNNs for solving the mathematical fractional order nonlinear economic and environmental system

$$\begin{cases} \frac{d^{(\rho)} u(\varepsilon)}{d\varepsilon^{(\rho)}} = C_3 w(\varepsilon) + C_1 u(\varepsilon)(a - u(\varepsilon)) - C_2 v(\varepsilon), & u_0 = k_1, \\ \frac{d^{(\rho)} v(\varepsilon)}{d\varepsilon^{(\rho)}} = C_4 u(\varepsilon)(a - u(\varepsilon)) + C_5 w(\varepsilon) - C_3 v(\varepsilon)(b - v(\varepsilon)), & v_0 = k_2, \\ \frac{d^{(\rho)} w(\varepsilon)}{d\varepsilon^{(\rho)}} = C_7 u(\varepsilon) - C_8 v(\varepsilon), & w_0 = k_3, \end{cases}$$

Mathematical model

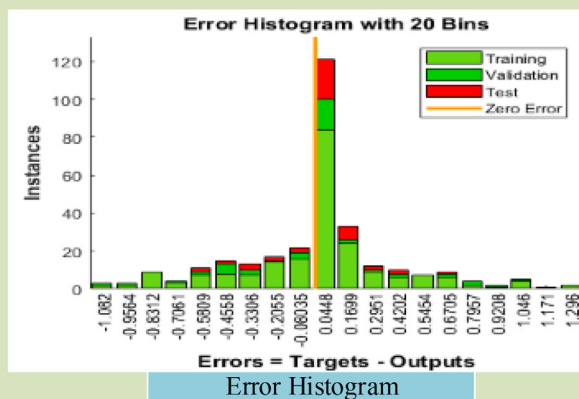
2. Methodology: SCGNNs

Reference solutions

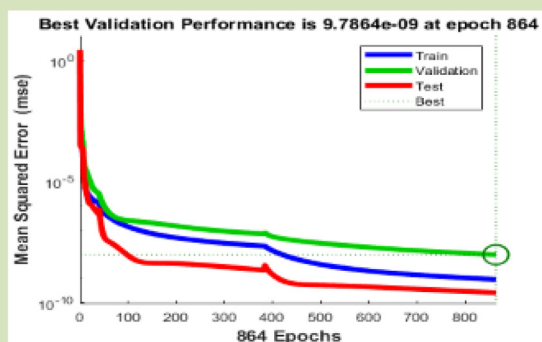
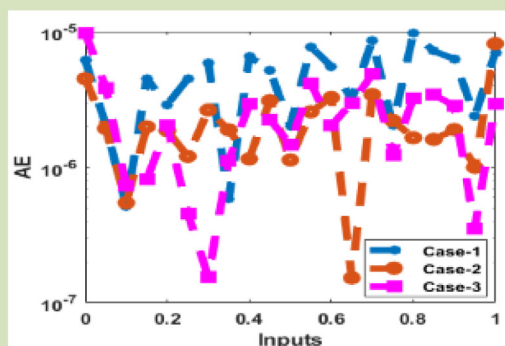
A dataset is designed by using the stochastic SCGNNs procedure for the mathematical fractional order nonlinear economic and environmental system

Proposed results

Activate the stochastic SCGNNs procedure using the reference statics to investigate the approximate outcomes of the mathematical fractional order nonlinear economic and environmental system



3. Results with analysis



Approximate stochastic SCGNNs procedure and investigation through the regression, EHS, fitness and MSE and for the mathematical fractional order nonlinear economic and environmental system

Fig. 1 SCGNNs performance to solve the nonlinear FO-EEMM.

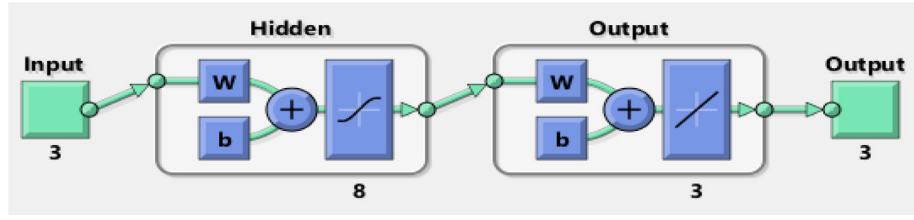


Fig. 2 Stochastic SCGNNs for the FO-EEMM.

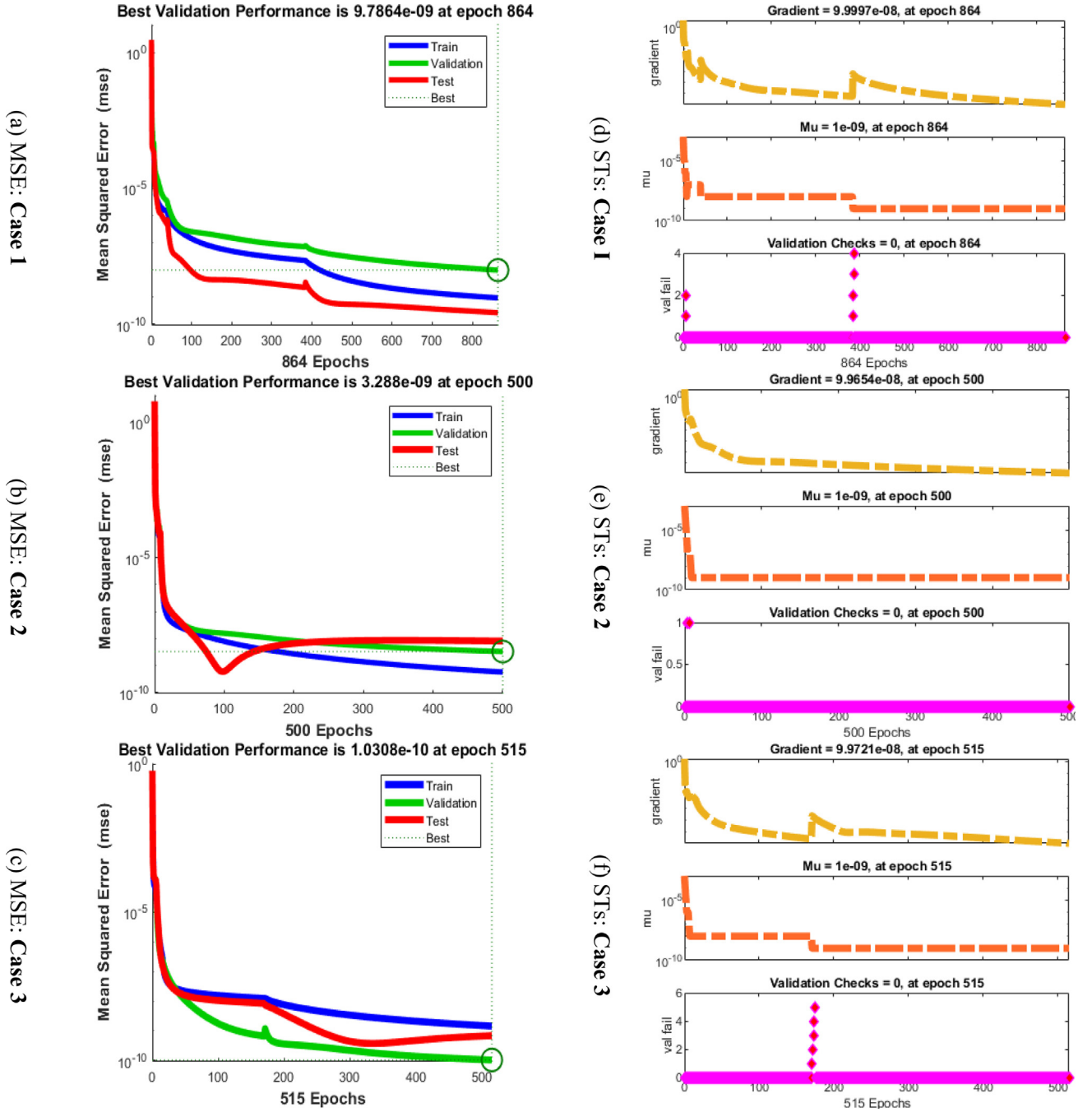


Fig. 3 MSE measures and the values of the STs for the FO-EEMM.

- By comparing the calculated and reference (Adams-Bashforth-Moulton) solutions, it is possible to determine whether the scheme is valid.
- The absolute error (AE) values are authenticated with the overlapping of 4 to 6 orders for the mathematical FO-EEMM.
- The constancy and reliability of the stochastic SCGNNs are established using the correlation, error histograms (EHs), regression, mean square error (MSE) and state transitions (STs) to indicate the numerical measures of the mathematical FO-EEMM.

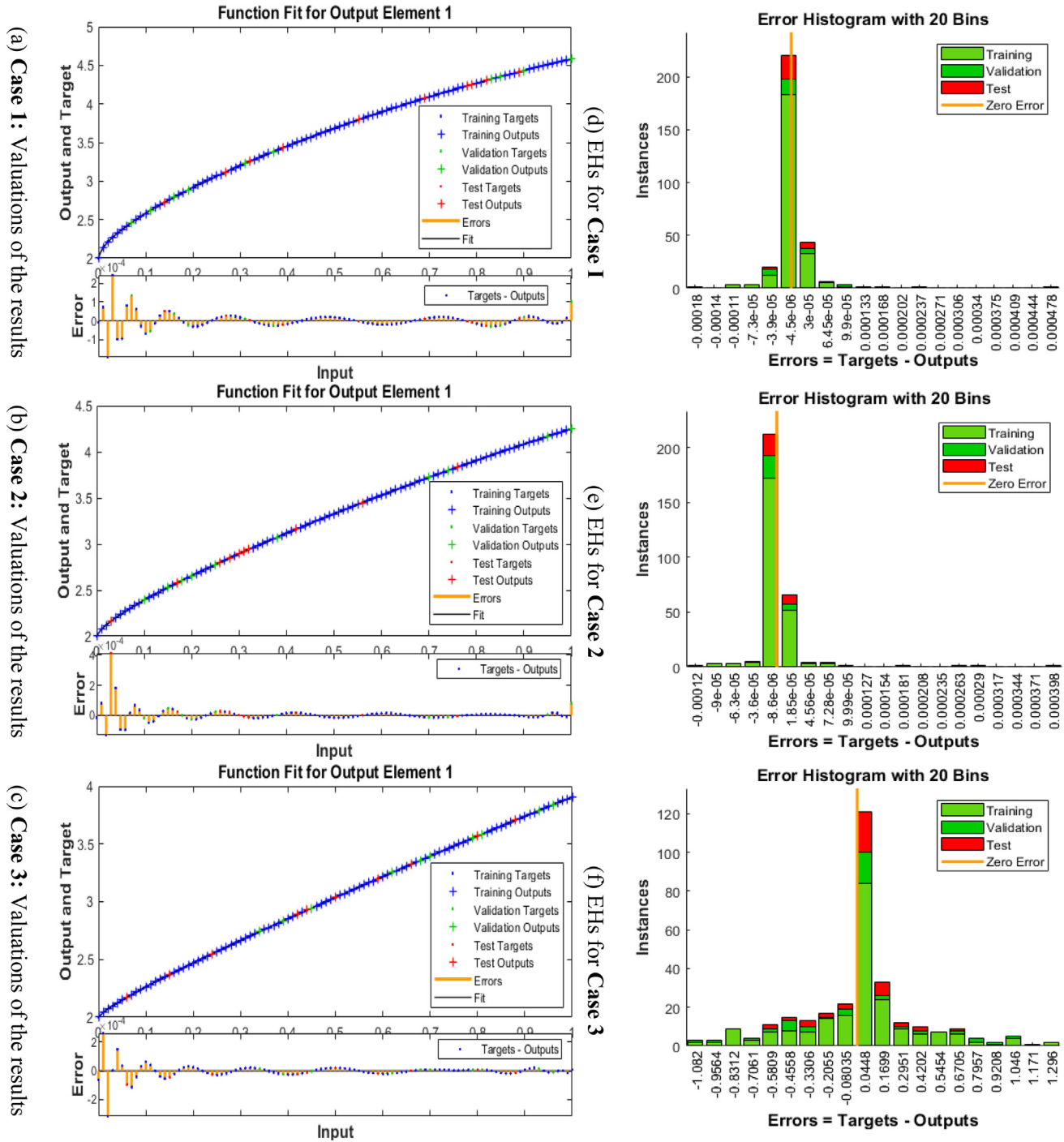
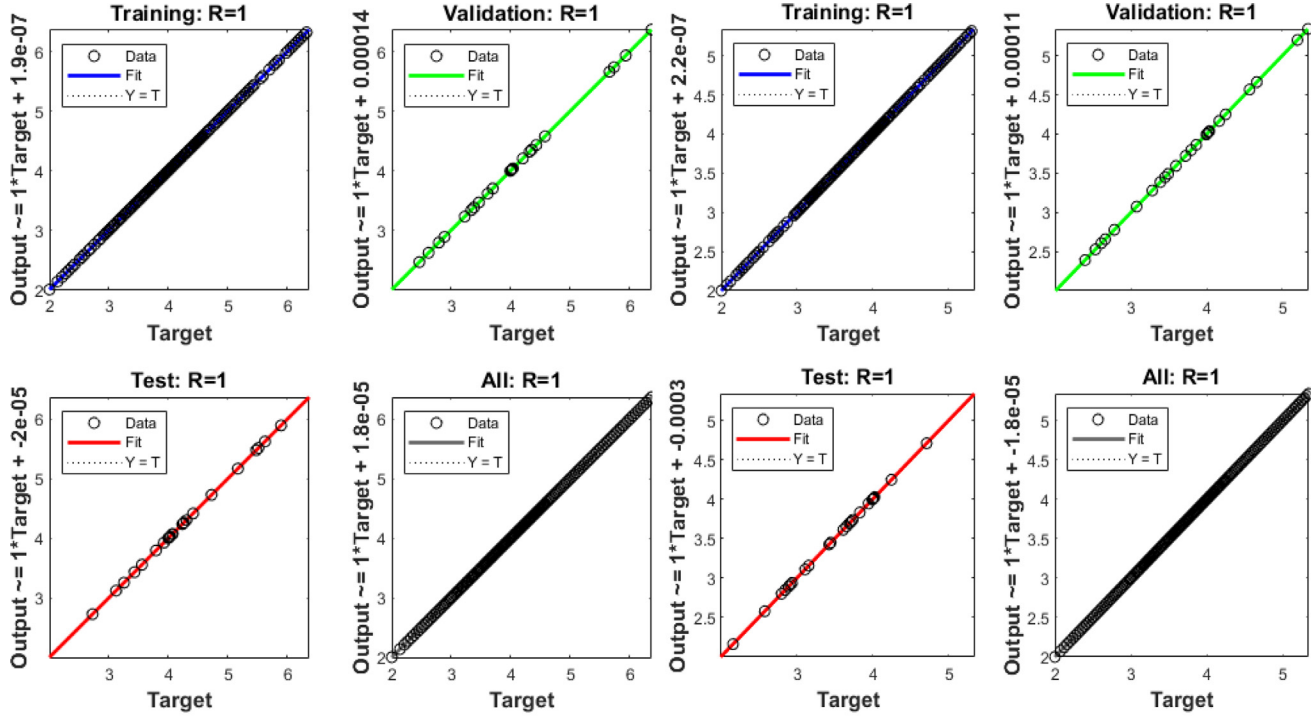


Fig. 4 EHs and evaluations of the results for the FO-EEMM.

3. Designed SCGNNs procedure

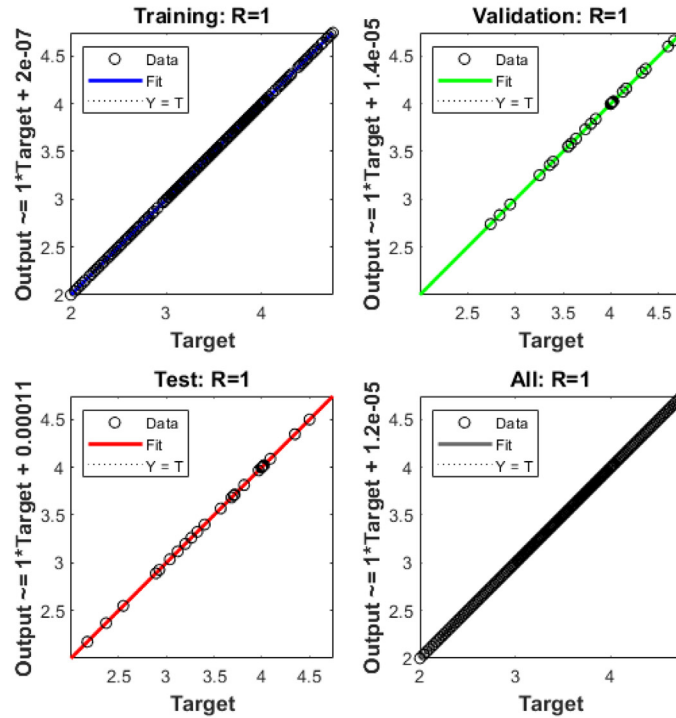
The current section shows the stochastic SCGNNs procedure to solve the mathematical FO-EEMM. The SCGNNs

approach is categorized as: the crucial measures based on the SCGNNs are given, whereas the accomplishment process is labeled to solve the nonlinear FO-EEMM. Fig. 1 presents the performances of the multi-layer optimization using the



(a) Regression values of the Case 1

(b) Regression values of the Case 2



(c) Regression values of the Case 3

Fig. 5 Regression estimations for the nonlinear FO-EEMM.

stochastic computing paradigms of the SCGNNs. The numerical performances are presented along with the selection of the statics as 76 % for training, 13 % for testing and 11 % for certification.

4. Numerical simulations

The current section indicates the investigations of the FO-EEMM. The mathematical representation of each variant is signified as:

Case 1. Suppose the FO-EEMM by using the suitable measures together with $\alpha = 0.6$ is written as:

$$\begin{cases} \frac{d^{0.6}u(\varepsilon)}{d\varepsilon^{0.6}} = 0.4w(\varepsilon) + 0.2(10 - u(\varepsilon))u(\varepsilon) - 0.3v(\varepsilon), u_0 = 2, \\ \frac{d^{0.6}v(\varepsilon)}{d\varepsilon^{0.6}} = 0.2u(\varepsilon)(10 - u(\varepsilon)) + 0.3w(\varepsilon) - 0.3v(\varepsilon)(5 - v(\omega)), v_0 = 3, \\ \frac{d^{0.6}w(\varepsilon)}{d\varepsilon^{0.6}} = 0.4u(\omega) - 0.3v(\varepsilon), w_0 = 4. \end{cases} \quad (3)$$

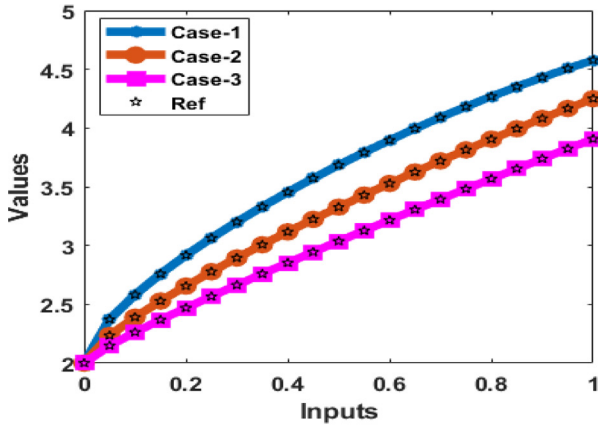
Case 2. Suppose the FO-EEMM by using the suitable measures together with $\alpha = 0.7$ is written as:

$$\begin{cases} \frac{d^{0.7}u(\varepsilon)}{d\varepsilon^{0.7}} = 0.4w(\varepsilon) + 0.2(10 - u(\varepsilon))u(\varepsilon) - 0.3v(\varepsilon), u_0 = 2, \\ \frac{d^{0.7}v(\varepsilon)}{d\varepsilon^{0.7}} = 0.2u(\varepsilon)(10 - u(\varepsilon)) + 0.3w(\varepsilon) - 0.3v(\varepsilon)(5 - v(\omega)), \\ v_0 = 3, \\ \frac{d^{0.7}w(\varepsilon)}{d\varepsilon^{0.7}} = 0.4u(\omega) - 0.3v(\varepsilon), w_0 = 4, \end{cases} \quad (4)$$

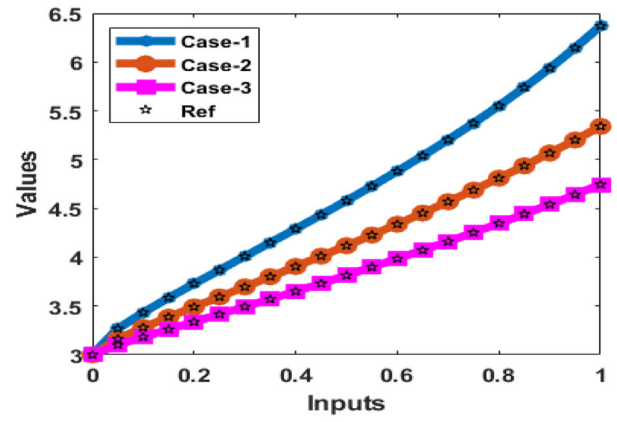
Case 3. Suppose the FO-EEMM by using the suitable measures together with $\alpha = 0.8$ is written as:

$$\begin{cases} \frac{d^{0.8}u(\varepsilon)}{d\varepsilon^{0.8}} = 0.4w(\varepsilon) + 0.2(10 - u(\varepsilon))u(\varepsilon) - 0.3v(\varepsilon), u_0 = 2, \\ \frac{d^{0.8}v(\varepsilon)}{d\varepsilon^{0.8}} = 0.2u(\varepsilon)(10 - u(\varepsilon)) + 0.3w(\varepsilon) - 0.3v(\varepsilon)(5 - v(\omega)), \\ v_0 = 3, \\ \frac{d^{0.8}w(\varepsilon)}{d\varepsilon^{0.8}} = 0.4u(\omega) - 0.3v(\varepsilon), w_0 = 4. \end{cases} \quad (5)$$

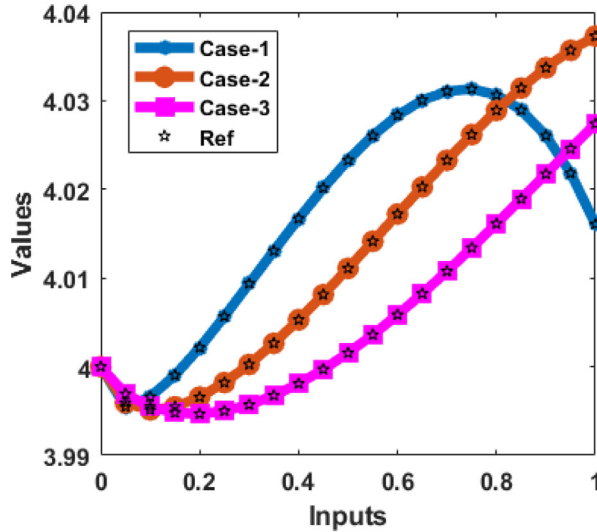
The stochastic computing performances of the FO-EEMM are signified by using the SCGNNs with 8 neurons together with the selection of the statics as 76 % for training, 13 % for testing and 11 % for certification. The output, input and hidden layer structure with 8 neurons are signified in Fig. 2.



(a) $u(\varepsilon)$ outcomes



(b) $v(\varepsilon)$ outcomes



(c) $w(\varepsilon)$ outcomes

Fig. 6 Result assessment for the FO-EEMM.

Figs. 3-7 represents the graphical diagrams for the mathematical FO-EEMM using the stochastic SCGNNs. The graphical depictions are shown in the Figs. 3 and 4 to authenticate the best performances and STs. The STs and MSE measures using the substantiation, best curves and training are illustrated in Fig. 3 for the FO-EEMM. The best presentations have been achieved by using the SCGNNs for the FO-EEMM at epochs 864, 500 and 515, which have been performed as 9.786×10^{-09} , 3.288×10^{-09} and 1.495×10^{-12} . The gradient values have been indicated by using the SCGNNs. These calculated performances found around 9.9997×10^{-08} , 9.9654×10^{-08} and 9.9721×10^{-08} . These illustrations indicate the SCGNNs convergence for the FO-EEMM. The values of the fitting curve are derived in Fig. 4 for each variant of the FO-EEMM based on the reference and achieved solutions. The error depictions have been illustrated through the testing, substantiation and training for the FO-EEMM. The EHs representations are derived in Fig. 4(a-c), whereas the regression is provided in Fig. 4(d-f).

The performances of the EHs have been performed as -4.50×10^{-06} , 1.85×10^{-05} and 1.60×10^{-06} for each case of the FO-EEMM. The correlation illustrations have been drawn in Fig. 5 to validate the performances of the regression. One can prove that correlation for each variant of the model is 1, which authenticates the perfect form of the modelling. The testing, verification and training representations designate the accurateness and precision of SCGNNs. The MSE values of the convergence have been proved to authenticate the training, complexity actions, generations, backpropagation, authentication, and testing, which is authenticated in Tab. 1 for the mathematical FO-EEMM.

Figs. 6 and 7 indicate the comparison measures through the matching of the results together with the assessment of the AE for the FO-EEMM. The numerical depictions for each category of the FO-EEMM have been illustrated by using the stochastic representations of the SCGNNs. The result depictions of the reference and proposed solutions are illustrated in Fig. 6, which signifies the overlapping of the solutions.

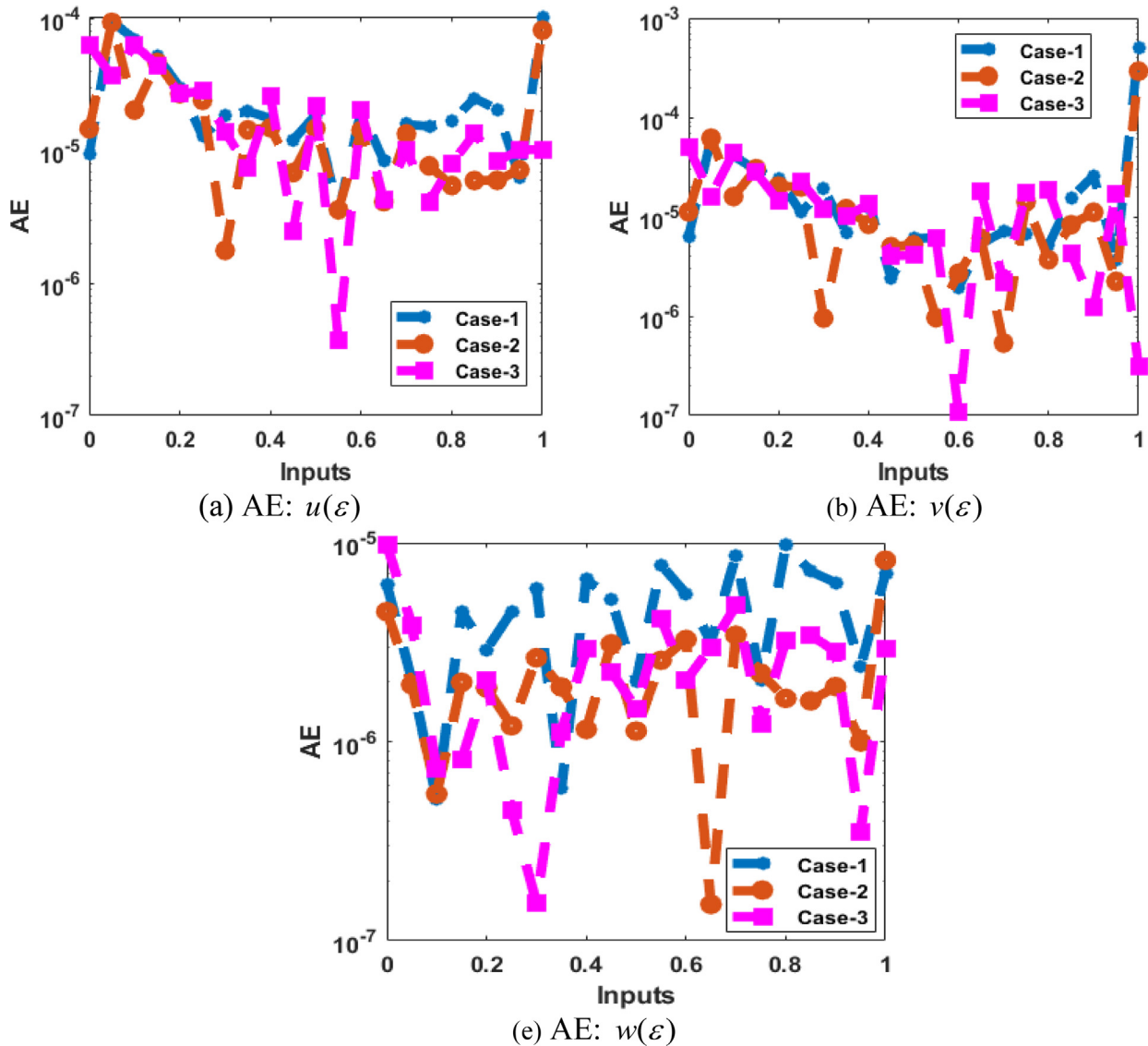


Fig. 7 Values of the AE for the FO-EEMM.

These precise matching validate the exactness of the computing SCGNNs to solve the FO-EEMM. The AE performances are illustrated in Fig. 7 for the mathematical FO-EEMM. The AE values for $u(\varepsilon)$ are provided as 10^{-04} to 10^{-05} , 10^{-04} to 10^{-06} and 10^{-04} to 10^{-07} for the respective variations of the FO-EEMM. The AE performances for $v(\varepsilon)$ are calculated as 10^{-04} to 10^{-06} , 10^{-05} to 10^{-07} and 10^{-04} to 10^{-07} for the respective variations of the FO-EEMM. The performances of the AE for $w(\varepsilon)$ lie as 10^{-05} to 10^{-06} , 10^{-05} to 10^{-07} and 10^{-06} to 10^{-07} for the respective variations of the FO-EEMM. These accurate AE represents the exactness of the scheme.

5. Conclusion

The aim of these investigations is to provide the numerical simulations based on the fractional order economic and environmental mathematical model. The construction of the FO-EEMM is categorized into three dynamics, control accomplishment cost, capability of the manufacturing elements and the diagnostics cost of the technical exclusion. Few concluding remarks are presented below as:

- To find more accurate solutions of the EEMM, experiments based on FO derivatives have been provided.
- The artificial neural network procedures along with the Scaled conjugate gradient have been used to present the FO-EEMM investigations numerically.
- Eight neurons have been selected along with the data as 76 % for training, 13 % for testing and 11 % for certification.
- The AE is performed in good measures and overlapping is performed through the comparison of the results, which shows the exactness of the scheme.
- The exactness, dependability and competence of the SCGNNs has been observed through the performances of the MSE, regression, STs, EHs and correlation.

In future, the SCGNNs approaches may be applied to solve the numerical representations of the nonlinear, fluid, biological systems and nonlinear models.

Declarations.

Author contributions.

All authors have read and agreed to the equal contribution of this manuscript.

Informed Consent Statement.

Not applicable.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] D.J. Reyniers, Supplier-customer interaction in quality control, *Ann. Oper. Res.* 34 (1) (1992) 307–330.
- [2] S. Whang, Coordination in operations: a taxonomy, *J. Oper. Manage.* 12 (3–4) (1995) 413–422.
- [3] C.R. Sox et al, Coordinating production and inventory to improve service, *Manage. Sci.* 43 (9) (1997) 1189–1197.
- [4] J.T. Mentzer et al, The nature of interfirm partnering in supply chain management, *J. Retail.* 76 (4) (2000) 549–568.
- [5] M.S. Mentzer, Two heads are better than one if your company spans the globe, *Academy of Management Perspectives* 13 (2) (1999) 89–90.
- [6] J. Mohr et al, Characteristics of partnership success: partnership attributes, communication behavior, and conflict resolution techniques, *Strateg. Manag. J.* 15 (2) (1994) 135–152.
- [7] H. Min et al, Supply chain modeling: Past, present and future, *Comput. Ind. Eng.* 43 (2002) 231–249.
- [8] Oliinyk, A et al., 2020. The use of the apparatus of ordinary differential equations in simulation of economic and environmental systems. In International scientific and technical conference Information technologies in metallurgy and machine building (pp. 216–220).
- [9] K. Nisar, Z. Sabir, M.A. Zahoora Raja, A.A.A. Ibrahim, S.R. Mahmoud, M. Balubaid, D.B. Rawat, J.J. Rodrigues, Numerical Study of the Environmental and Economic System through the Computational Heuristic Based on Artificial Neural Networks, *Sensors* 21 (19) (2021) 6567.
- [10] A.K. Kiani, W.U. Khan, M.A.Z. Raja, Y. He, Z. Sabir, M. Shoaib, Intelligent backpropagation networks with bayesian regularization for mathematical models of environmental economic systems, *Sustainability* 13 (17) (2021) 9537.
- [11] V.S. Panwar, P.S.S. Uduman, Existence and Uniqueness of Solutions for Mixed Immunotherapy and Chemotherapy Cancer Treatment Fractional Model with Caputo-Fabrizio Derivative, *Progress in Fractional Differentiation and Applications An International Journal* 8 (2) (2022) 243–251.
- [12] A. Aghili, Complete Solution For The Time Fractional Diffusion Problem With Mixed Boundary Conditions by Operational Method, *Applied Mathematics and Nonlinear Sciences* 6 (1) (2021) 9–21.
- [13] H. Bulut, A.N. Akkilic, B.J. Khalid, Soliton solutions of Hirota equation and Hirota-Maccari system by the $(m+1/G')$ -expansion method, *Advanced Mathematical Models and Applications* 6 (1) (2021) 22–30.
- [14] Begum, R., Tunç, O., Khan, H., Gulzar, H. and Khan, A., 2021. A fractional order Zika virus model with Mittag-Leffler kernel. *Chaos, Solitons & Fractals*, 146, p.110898.
- [15] Bulut H., Ismael, H.F., 2022. Exploring New Features for The Perturbed Chen-Lee-Liu Model Via $(M+1/G')$ -Expansion Method, *Proceedings of the Institute of Mathematics and Mechanics, National Academy of Sciences of Azerbaijan*, 48 (1), pp:164–173.
- [16] A. Jajarmi, D. Baleanu, K. Zarghami Vahid, S. Mobayen, A general fractional formulation and tracking control for immunogenic tumor dynamics, *Mathematical Methods in the Applied Sciences* 45 (2) (2022) 667–680.
- [17] V.S. Erturk, E. Godwe, D. Baleanu, P. Kumar, J. Asad, A. Jajarmi, Novel fractional-order Lagrangian to describe motion of beam on nanowire, *Acta Phys. Pol. A* 140 (3) (2021) 265–272.
- [18] A. Jajarmi, D. Baleanu, K.Z. Vahid, H.M. Pirouz, J.H. Asad, A new and general fractional Lagrangian approach: a capacitor microphone case study, *Results Phys.* 31 (2021).
- [19] A. Alkhazzan, P. Jiang, D. Baleanu, H. Khan, A. Khan, Stability and existence results for a class of nonlinear fractional differential equations with singularity, *Mathematical Methods in the Applied Sciences* 41 (18) (2018) 9321–9334.
- [20] M. Gürbüz, C. Yıldız, Some new inequalities for convex functions via Riemann-Liouville fractional integrals, *Applied Mathematics and Nonlinear Sciences* 6 (1) (2021) 537–544.
- [21] A. Kumar, A. Prakash, H.M. Baskonus, The epidemic COVID-19 model via Caputo-Fabrizio fractional operator, *Waves Random Complex Medium* (2022), <https://doi.org/10.1080/17455030.2022.2075954>; pp1–12.

- [22] B.M.H., Flora, S.A., On the analytical approximation of the quadratic non-linear oscillator by modified extended iteration method, *Applied Mathematics and Nonlinear Sciences*, 6(1), pp. 527 – 536, 2021.
- [23] P. Veeresha, E. Ilhan, D.G. Prakasha, H.M. Baskonus, W. Gao, A new numerical investigation of fractional order susceptible-infected-recovered epidemic model of childhood disease, *Alexandria Eng. J.* 61 (2) (2022) 1747–1756.
- [24] E. Ilhan, P. Veeresha, H.M. Baskonus, Fractional approach for a mathematical model of atmospheric dynamics of CO₂ gas with an efficient method, *Chaos, Solitons Fractals* 152 (111347) (2021) 1–10.
- [25] H. Ismael, H.M. Baskonus, H. Bulut, Abundant novel solutions of the conformable Lakshmanan–Porsezian–Daniel model, *Discr. Continuous Dynam. Syst.-S* 14 (7) (2021) 2311–2333.
- [26] S. Momani, I.M. Batiha, Tuning of the Fractional-order PID Controller for some Real-life Industrial Processes Using Particle Swarm Optimization, *Progress in Fractional Differentiation and Applications An International Journal* 8 (3) (2022) 377–391.
- [27] H. Jun, L.C. Xi, X. Chun-Long, L. Wei-Xiong, Y. Jun-Jian, A. Lozano, Decision-making system and verification of pavement diseases treatment scheme for highway reconstruction and extension, *Appl. Math. Nonlinear Sci.* 6 (1) (2021) 151–162.